Evoqed Livestock — Product guide

Prepared: Monday 2 November 2026 (Nepal)
Live web: https://evoqedlivestockfarm.vercel.app
Repo: github.com/manish3321/Livestock · Mobile: Expo / EAS preview APK
Evoqed Livestock (also called Kharka Farm OS in older docs) is a farm operating system for commercial and mixed farms in Nepal. It covers the shed, breeding desk, health, batches, inventory, money, and offline phone work in one shared API.

1. What it is

- One NestJS API (/v1) as source of truth, with farm-scoped tenancy on every row.
- React web dashboard for owners, managers, workers, vets, and co-ops.
- Expo React Native phone app for shed rounds, breeding Watch/Walk/Herd, inbox, and offline sync.
- English and Nepali ((G*>2@) on web and phone.
- Postgres (Supabase), JWT auth, push/SMS/voice notification pipeline for Inbox tasks.

2. Who uses it (roles)

- ADMIN — full access: ops, money, reports, members, audit.
- MANAGER — operations plus approvals, revenue, P&L, reports, change approvals.
- WORKER — data entry and shed/rounds; Watch + Walk only in Breeding (no Herd KPIs).
- VET — animals (read), health/breeding write, inventory read, rounds; no stock write or money.
- COOP — read animals/production/revenue/reports; no Shed or Breeding write.

Details: docs/rbac.md. The API enforces permissions even if the UI shows a button.

3. Main modules

Home (dashboard)

Greeting, yesterday’s milk/eggs/fish, urgent alerts (overdue health, low stock), quick links, and today’s Inbox tasks. Needs a healthy /v1/dashboard/summary call.

Shed

- Modes: Milking (Morning / Evening / Midday), Vaccination, Treatment, Weighing, Health check, Band (markers).
- Start or resume a recording round; type herd number / tag or tap remaining animals.
- Blank litres stay disabled; huge litres rejected; unknown codes do not attach to another animal.
- Finish with skip reasons (including DRIED_OFF); abandon marks ABANDONED without inventing a tank total.
- Tank and delivery cannot exceed milked SOLD litres (TANK_ABOVE_MILKE
a HIGH tank shortfall task.
- Daily sheet shows the session’s animals, skips, and totals.

Breeding

- Watch — planning windows (heat, breed today, dry-off, calving, colostrum).
- Walk — today’s actionable desk.
- Herd — KPIs (conception, days open, heat detection, observer table). Managers/admins/vets only — not workers.
- Sheets: heat, service, pregnancy check (PD), calving, colostrum, dry-off.
- Species-aware windows (buffalo vs cow); goats are not offered cattle protocols.

Livestock & batches

- Animal picker counts by species in one request; open a species for the paged list.
- Batch detail loads economics, illness, mortality, feed, water, sampling, harvest in one combined payload (inventory picker can stay separate).

Health, inventory, feed, production, money

- Vaccinations/treatments pick inventory; stock cannot go negative without STOCK_NEGATIVE.

- Milk withhold markers show on the milking pad; withhold blocks SOLD milk.
- Expenses: worker submits ’ manager approves ’ large amounts escalate to
- Revenue, P&L, reports for roles that have finance/reports permissions.

Inbox & notifications

- Every reminder is a farm task. Dispatcher sends push; CRITICAL may also SMS/voice.
- Anti-fatigue: night hold for non-urgent, daily caps, bundling, escalation if unacked.
- See docs/notification-reminders.html (updated 2 Nov 2026).

4. Phone & offline

- Bottom nav: Home, Shed, Scan QR, Inbox, More.
- Offline module cache pages past 100 rows (cap 20 pages per module) so large herds hydrate fully.
- Outbox sync: failed mutations stay failed/pending — never silently marked success as duplicates.
- EXPO_PUBLIC_API_URL points at production (EAS preview) or http://10.0.

5. Technical shape

- Monorepo: apps/api (Nest), apps/web (Vite React), apps/mobile (Expo 52), packages/contracts + design-tokens.
- Local API may use DIRECT_URL (session :5432) for fast shed reads; Vercel must use pooler DATABASE_URL.
- Deploy: Vercel hosts API+web; mobile via EAS Build (preview APK).

6. How to sign in locally

- Seed users (password from SEED_PASSWORD, default ChangeMe123!): owner/manager/worker@farm.local; test VET/COOP if seeded.
- Web :5173 · API :4001 · Metro :8081.

— End of product guide —